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- Why are biological systems modeled as networks? (A) To reduce data size (B) To capture interactions between entities (C) To avoid experiments (D) To simplify visualization only
- Nodes in biological networks typically represent: (A) Only genes (B) Only proteins (C) Biological entities (genes, proteins, et(C)) (D) Only pathways
- Edges in biological networks represent: (A) Physical distance (B) Interactions or relationships (C) DNA sequences (D) Experimental errors
- Which network is typically undirected? (A) Gene regulatory network (B) Metabolic network (C) Protein-protein interaction network (D) Signaling network
- Directed edges are most suitable for: (A) PPIs (B) Random graphs (C) Gene regulation (D) Social networks only
- Scale-free networks are characterized by: (A) Equal degree distribution (B) Presence of hubs (C) No connections (D) Fully connected nodes
- Main advantage of scale-free networks: (A) Easy visualization (B) Robust to random failures (C) No hubs (D) No clustering
- Main weakness of scale-free networks: (A) Random failure (B) Targeted attack on hubs (C) Too many edges (D) Low clustering
- Small-world property implies: (A) Large distances between nodes (B) Very few nodes (C) Short path lengths across network (D) No clustering
- Clustering coefficient measures: (A) Node degree (B) Path length (C) Tendency of nodes to form clusters (D) Edge weights
- A node has degree $k = 3$ and 3 connections among neighbors. The clustering coefficient is: (A) 0.33 (B) 0.50 (C) 1.00 (D) 0.67
- In a network, a node has degree $k = 5$. The maximum possible number of edges among its neighbors is: (A) 5 (B) 10 (C) 15 (D) 20
- If the shortest distances from node i to all other nodes are: 2, 2, 4, 2, what is the closeness centrality (unnormalized inverse)? (A) $\frac{1}{10}$ (B) $\frac{1}{8}$ (C) $\frac{1}{6}$ (D) $\frac{1}{4}$
- Betweenness centrality measures: (A) Degree (B) Clustering (C) Control over information flow (D) Node similarity
- Modules (communities) are: (A) Random nodes (B) Highly disconnected nodes (C) Densely connected groups (D) Isolated nodes